

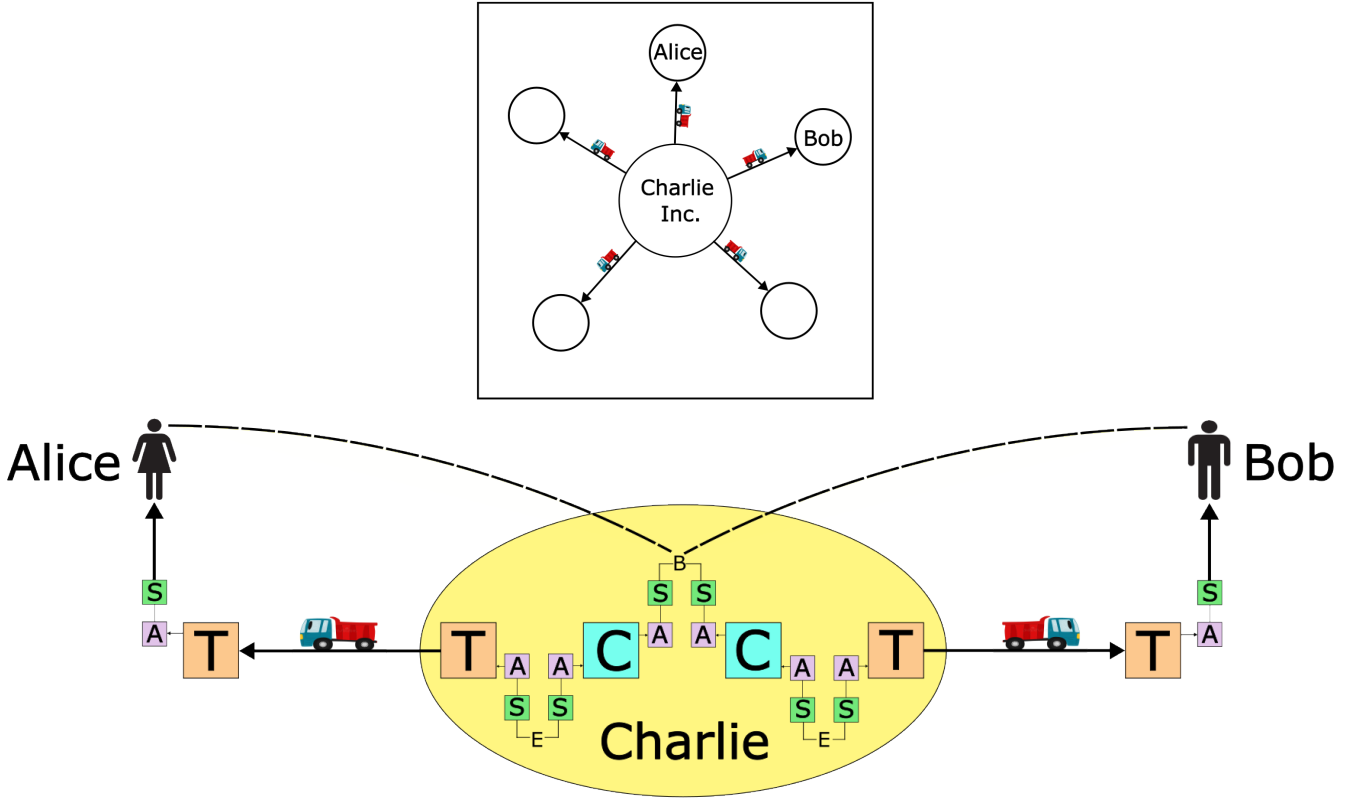


Figure 2. (Above) A schematic of a 5-node delayed choice quantum network with Charlie Inc. in the center. (Below) Delayed choice entanglement swapping protocol between two nodes Alice and Bob. S represents the surface code patches, A refers to the ancilla patches encoded in HGP code, C denotes Charlie’s qLDPC memory and T represents the qLDPC memory transported to the end-nodes. Here, E represents the initial entangled Bell-pair and B denotes bell measurement.

plication, sacrificing some bits in the process. The remaining bits form a secure encryption key for end-to-end message encryption. This approach turns entanglement into a tradeable commodity and allows users with only classical hardware to participate in quantum cryptography protocols. For example, Alice measures a number of qubits, labeled $A_1, A_2, \dots, A_n$ and informs Charlie Inc. which qubit measurement results she now owns. Charlie Inc. then records the owner of the qubits and stores the other half of the Bell pairs in long-term quantum storage. Similarly, Bob buys the rights to measure qubits $B_1, B_2, \dots, B_n$ and Dave buys the rights to measure qubits $D_1, D_2, \dots, D_n$. At this stage, neither participant has decided who they wish to exchange correlations with, they have simply ‘topped up’ their future quantum potential i.e. the bits with which they can perform an unconditionally secure one-time-pad with another user of their choice in the future. If Alice and Dave wish to generate entangled data for some protocol, then they need to publicly declare to Charlie Inc. how many topped up bits they want to have correlated, then Charlie Inc. performs the entanglement swap between the appropriate (stored) Bell pairs and broadcasts the measurement results. If Alice and Dave wish to check Charlie Inc.’s fidelity, they may sacrifice some of their bits, and assuming they are

satisfied, can proceed with standard key distillation and privacy amplification [39]. If Alice then wishes to share secrets with Bob, then again, she need only declare to Charlie Inc. how many bits she wants to correlate (of course, she cannot use measurement results that have previously been used with Dave), and proceed as before.

IV. RESOURCE ESTIMATION

In this section, we derive closed-form expressions for logical failure rates, time scales, and logistics concerning the protocol. For our analysis, we consider city-wide and district-wide distances, hence we will be considering trucks or train cars for transport of quantum memories. We consider both trucks and train cars to be equivalent in carrying capacity, transport speeds, etc., and may tend to use the terms interchangeably. Throughout this section, we consider the processes and operations occurring in the quantum error-corrected qLDPC memories as they go through all the stages (loading, transport, and unloading) of the protocol.

For our protocol, we select the quantum LDPC codes constructed from the hypergraph product of classical codes based on (3, 4)-regular bipartite graphs with strong

expansion properties [27]. These graphs are selected randomly, and the resulting quantum codes exhibit a minimum encoding rate of $1/25$. In ref. [27], the authors discuss the implementation of computation with qLDPC codes using neutral atoms, which provide an order of magnitude improvement in the number of physical qubits used compared to surface codes. They introduce a technique by which logical qubits can be teleported between the surface codes and qLDPC code blocks, thus aiding in the possibility of computation and individual logical qubit measurements, which becomes essential for the implementation for our protocol. The authors also provide closed-form expressions for various parameters such as logical failure rates and cycle times, which we incorporate in our resource analysis.

Rearrangement of atoms: For a $[[n, k, d]]$ hypergraph product (HGP) code with a coding rate $r = k/n$ where n is the number of physical qubits, k is the number of logical qubits, and d is the code distance, each syndrome extraction (SE) cycle takes time $8t_r$ [27], where the factor 8 represents the number of atomic rearrangements needed to complete one error correction cycle, and t_r is the time required to move the atomic qubits around for one rearrangement [27]. By overlapping the syndrome measurements with the atom repositioning of the next iteration, we can avoid extending the total execution time. We express t_r as a function of n , given by:

$$t_r(n) \approx 2\tau_t \log L + (3 + 2\sqrt{2})\sqrt{\frac{6Ld_p}{a_p}} \quad (1)$$

where $\tau_t = 50\mu s$ is the transfer time between the atomic traps, $a_p = 0.02\mu m\mu s^{-2}$ is the maximum acceleration rate of the atom, $d_p = 5\mu m$ is the spacing between the atoms in the qLDPC grid, and $L \approx \sqrt{n}$ is the number of atoms in a line to be rearranged. The gate times of neutral atom architectures are in the order of microseconds [40] while t_r is in the order of several milliseconds. Therefore t_r becomes the dominant rate-limiting time scale and the gate times can be ignored while computing neutral atom qLDPC cycle times.

Logical Failure Rates (LFRs): The Logical Failure Rate (LFR) R_L per error correction cycle for a $[n, k = n/25, d = \sqrt{k}]$ HGP code is given by [27]:

$$R_L(n) = 0.07(p_g/0.006)^{0.47}n^{0.27} \quad (2)$$

where p_g is the physical gate error and n is the number of physical qubits, $k = n/25$ is the number of logical qubits, and $d = \sqrt{k}$ is the code distance. Ref[27] considers HGP codes such that $k = n/25$ and $d = \sqrt{k} = \sqrt{n}/5$. We consider this configuration as well in our analysis. Similarly, the Logical Failure Rate (LFR) R_S per error correction cycle for a surface code of distance d_{sc} is given by [21]:

$$R_S = 0.3(70p_g)^{\frac{d_{sc}+1}{2}} \quad (3)$$

Let n_a be the number of physical qubits in the ancilla HGP code. Let n_t be the number of physical qubits in the

qLDPC memory to be transported. Let n_c be the number of physical qubits in the qLDPC memory of Charlie Inc.. We make two key assumptions in our analysis. First, we consider the number of physical qubits in Charlie Inc.'s qLDPC memory (n_c) and the transported qLDPC memory (n_t) to be equal, denoting both as n_m , where 'm' stands for memory. Secondly, we consider the distance of the computational surface code (d_{sc}) as a function of the qLDPC memory size, specifically $d_{sc} = \sqrt{n_m}/5$, as used by the analysis in [27].

Throughout our derivations, we use the following binomial approximation:

$$\begin{aligned} (1-p) &= (1-p_1)^{n_1}(1-p_2)^{n_2}\dots \\ \Rightarrow p &\approx n_1p_1 + n_2p_2 + \dots \\ \forall 0 &< p_i < 1 \end{aligned} \quad (4)$$

This approximation is valid due to the logical failure rates being much lower than 1, due to which logical errors can be propagated via simple addition.

Initialization of qLDPC code patches: First, Charlie Inc. needs to initialize two qLDPC patches, one (C) for his memory and the other (T) for transporting to the destination node. The Logical Failure Rate (LFR) R_0 for this is given by:

$$R_0 = 2\frac{\sqrt{n_m}}{5}R_L(n_m) \quad (5)$$

Here $d = \sqrt{n_m}/5$ is the code distance, which is also the number of error-correction cycles, and $R_L(n_m)$ is the LFR per error correction cycle. We assume that these memories are initialized in parallel with the previous memory batch being loaded onto the truck. Therefore, we do not count it into our time scales.

Bell Pair Creation: Next step, we need to create a Bell-pair. Now, For creating a single Bell Pair, we need two surface codes and $2\frac{\sqrt{n_m}}{5}$ error-correction cycles. The Logical Failure Rate (LFR) R_1 for this is given by [21, 41]:

$$R_1 = 2\frac{\sqrt{n_m}}{5}R_S \quad (6)$$

Since loading the current bell pair and creating the next bell pair are done simultaneously, we don't count this time scale.

Loading of the Bell Pair: For the loading of both halves of the surface-coded Bell-Pair onto their respective qLDPC patches C and T , we need to teleport it from the surface code to the HGP patch via an ancilla HGP. The ancilla HGP is of the configuration $[[n_a = n_m/25, k = 1, d = \sqrt{n_a}]]$. For the teleportation, we need first to initialize the ancillae. then we need to do two lattice surgery procedures, one merge-and-split between the surface code and the ancilla HGP patch, and another between the ancilla HGP and the qLDPC patches. the

LFR for loading onto the transport qLDPC is given by:

$$R_2 = 2(\sqrt{n_a}R_L(n_a) + \frac{\sqrt{n_m}}{5}(R_S + R_L(n_a)) + \frac{\sqrt{n_m}}{5}(R_L(n_a) + \frac{n_m}{25}R_L(n_m))) \quad (7)$$

where the first term is for the ancilla initialization involving $d = \sqrt{n_a}$ rounds of error correction, the second term is for lattice surgery between surface code and ancilla involving $d = \frac{\sqrt{n_m}}{5}$ error correction cycles, with LFR per cycle for the surface code being R_S and that of the ancilla HGP being $R_L(n_a)$, and the third term is for that of ancilla and the memory patch. The patch sizes are chosen in such a way that all three patches (memory, ancilla, and the surface code) have the same code distance, and hence require the same number of error correction cycles. Here the $\frac{n_m}{25}$ factor appears since the first loaded qubit will undergo these sets of loading QEC cycles $\frac{n_m}{25}$ times, which propagates its error. The term 2 arises because we need to teleport two halves of the Bell-pair, one to the transport memory T and the other one to Charlie Inc.'s memory C . The time T_2 required for this is given by:

$$T_2 = \frac{n_m}{25} \left(\sqrt{n_a}t_{cyc}(n_a) + \frac{\sqrt{n_m}}{5}t_{cyc}(n_a) + \frac{\sqrt{n_m}}{5}t_{cyc}(n_m) \right) \quad (8)$$

where $t_{cyc} = 8t_r$ is the time taken to complete one error correction cycle. Here we combine parallel cycles, such as that happening simultaneously on surface code and ancilla, and on the ancilla and the qLDPC memory.

Storage and Transport: Let's say the destination node is T_3 units of time drive from Charlie Inc.. Throughout the transit, $M_t = \frac{T_3}{T_{cyc}(n_m)}$ number of error correction cycles happen on the transport memory T . The LFR R_3 occurring during the transport is given by:

$$R_3 = \frac{T_3}{T_{cyc}(n_m)} R_L(n_m) \quad (9)$$

Here, the idling errors, which emerge as an adjustment to the value of p_g , have been incorporated into R_L , and that a scenario has been considered where it is beneficial to perform QEC cycles as often as possible. This is explained in section V under the topic of idling errors (Eq.20).

Unload and Measure at qATM: Once the transport memory arrives at a qATM in the destination node, we need to initialize the surface code and the ancilla, and then sequentially teleport the logical qubits from the qLDPC to the surface code via the ancilla, and measure

the surface code qubit. The LFR for this is given by:

$$R_4 = \frac{\sqrt{n_m}}{5}R_S + \sqrt{n_a}R_L(n_a) + \frac{\sqrt{n_m}}{5}(R_S + R_L(n_a)) + \frac{\sqrt{n_m}}{5}(R_L(n_a) + \frac{n_m}{25}R_L(n_m)) + R_S \quad (10)$$

This is similar to Eq. 7, with an additional term R_S in the end which is the 1-round LFR that occurs due to measurement. The time taken for this is:

$$T_4 = \frac{n_m}{25} \left(6\frac{\sqrt{n_m}}{5}t_g + \sqrt{n_a}t_{cyc}(n_a) + \frac{\sqrt{n_m}}{5}t_{cyc}(n_a) + \frac{\sqrt{n_m}}{5}t_{cyc}(n_m) + 6t_g \right) \quad (11)$$

Measuring a surface-code logical qubit involves one round of error correction. The depth of a surface-code syndrome extraction is 6 and the physical gate time of the architecture is given by t_g . Hence the last term.

Now, the total network LFR R_{net} of the processes so far, and the total time T_{tot} taken for generating this correlation is:

$$R_{net} = \sum_{j=0}^4 R_j, \quad T_{tot} = \sum_{j=2}^4 T_j \quad (12)$$

Unload and Bell-measure at Charlie Inc.'s end:

Now we have two such half-measured Bell-Pairs distributed between the network pairs Charlie Inc.-Alice and Charlie Inc.-Bob. So we have two memory drives, one each for each network pair. Let's say at a later time T_{later} , Alice and Bob decide to correlate their bits. For calculation purposes, we assume that this decision has been made by the time $T_{later} = T_{tot}$. We assume that by this time, the end users Alice and Bob have decided to communicate with Charlie Inc. and have informed him accordingly. Let the total number of error correction cycles M_{tot_c} be done by Charlie Inc.'s memory drives during this time. This is given by $M_{tot_c} = T_{later}/t_{cyc}(n_m)$. Since we have two drives, the LFR R_{store} is given by:

$$R_{store} = 2M_{tot_c}R_L(n_m) = 2\frac{T_{tot}}{t_{cyc}(n_m)}R_L(n_m) \quad (13)$$

The factor of 2 occurs because there are two memory drives. Now the qubits need to be teleported onto surface